Algorithmic Pricing and Market Conduct: Evidence from Airbnb Smart Pricing

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Abstract

We study the market-level effects of Airbnb's redesigned algorithmic pricing tools across eight major U.S. cities. Using a sharp reduced-form regression discontinuity design exploiting the platform-wide September 2023 rollout, we estimate the intent-to-treat effect of algorithm availability on listing prices and price dispersion. The aggregate price-level effect is small and bandwidth-sensitive: the pooled ITT ranges from 1.9 percent at ±30 days to 0.5 percent at ±90 days — an order of magnitude below the effects documented in oligopolistic markets. A balanced-panel specification and controls for simultaneous platform policy changes confirm that the result is not driven by compositional shifts or bundled treatments. The RDD finds increased within-listing price variance, but a year-over-year DiD yields the opposite sign, leaving the variance result's robustness uncertain. Pricing changes concentrate among hosts with high pre-existing pricing sophistication, a descriptive pattern consistent with technology-skill complementarity. Together with evidence of algorithmic collusion in gasoline duopolies (Assad et al. 2024), these findings suggest that market structure mediates whether pricing algorithms raise prices or merely alter pricing patterns.

JEL codes: L13, L41, L81, D43, C21

Keywords: algorithmic pricing, price discrimination, regression discontinuity, platform markets, Airbnb, technology-skill complementarity

1 Introduction

Pricing algorithms are reshaping competition in digital marketplaces. In 2024, the U.S. Department of Justice sued RealPage for allegedly facilitating rent coordination through a shared algorithmic pricing tool used by competing landlords. The Federal Trade Commission launched a broad investigation into "surveillance pricing" across multiple industries. In Europe, the Digital Markets Act imposed new obligations on platform gatekeepers partly motivated by concerns about algorithmic market power. These enforcement actions reflect a central anxiety: that when competing sellers delegate pricing to algorithms — especially algorithms provided by a common platform — the result may be tacit collusion, even without explicit communication.

The theoretical case for concern is well-developed. Calvano et al. (2020) demonstrate that Q-learning agents converge to supra-competitive prices with implicit punishment strategies. Brown and MacKay (2025) show that a single firm with a fast, committed pricing algorithm can unilaterally coerce rivals into sustaining prices above the collusive benchmark. The empirical evidence, however, remains concentrated in settings maximally conducive to coordination: homogeneous products, duopoly or tight oligopoly, and real-time price transparency. Whether these findings generalize to the large, differentiated-product platform markets where algorithmic pricing is most prevalent remains an open question.

We study the rollout of Airbnb's redesigned pricing tools across eight major U.S. cities: Austin, Boston, Chicago, Los Angeles, New York City, San Francisco, Seattle, and Washington, DC. The platform's 2023 Summer Release introduced updated algorithmic pricing recommendations, enhanced dashboards, and comparison tools (Ye et al. 2018). Using a sharp reduced-form regression discontinuity design that exploits the platform-wide rollout on September 1, 2023, we estimate the intent-to-treat effect of algorithm *availability* on listing prices and price dispersion. Our estimation sample comprises 7.5 to 22.2 million listing-day observations (depending on bandwidth) across eight cities, with seasonal confounders removed via cross-fitted residualization following Hausman and Rapson (2018) and Noack et al. (2025).

Three findings emerge. First, the aggregate price-level effect is small and bandwidth-sensitive. The pooled ITT estimate on residualized log prices ranges from 1.9 percent at the narrowest bandwidth ($\$\pm 30 days) to 0.5 percent at the widest (\pm\$90 days), declining monotonically. All estimates are statistically significant but economically modest — far below the magnitudes associated with algorithmic collusion in oligopolistic markets. The attenuation with bandwidth is consistent with either a genuine local effect concentrated near the rollout date or residual seasonal confounding that inflates narrow-bandwidth estimates. A balanced-panel specification restricted to listings present on both sides of the cutoff produces nearly identical

estimates, indicating that compositional changes at the cutoff do not drive the result. Controlling for simultaneous changes in minimum-night requirements also leaves the estimates unchanged.

Second, the RDD finds that algorithm availability increases within-listing temporal price variance. The OLS local-linear estimate for 7-day rolling variance of log prices is 0.0074 ($p < 0.001$); for 14-day rolling variance, 0.0094 ($p < 0.001$). However, a year-over-year DiD specification yields variance estimates of the *opposite sign* (\$-\$0.0008), indicating that the variance finding is sensitive to the identification strategy. We present both results and discuss what must be true about secular trends for each to be the correct causal estimate.

Third, pricing changes concentrate among hosts with high pre-existing pricing sophistication. Using an unsupervised machine learning proxy for latent adoption propensity constructed from strictly pre-treatment pricing behavior, we find that high-propensity hosts exhibit significantly greater post-rollout variance increases. This descriptive pattern is consistent with technology-skill complementarity, though it may also reflect selection into pricing sophistication rather than differential treatment effects.

This paper contributes to three strands of literature. First, we contribute to the growing empirical literature on algorithmic pricing and competition. Assad et al. (2024) provide the strongest quasi-experimental evidence of algorithmic collusion in German gasoline duopolies. Musolff (2022) documents tacit coordination among Amazon repricing algorithms. Brown and MacKay (2025) establish theoretically that algorithms can raise prices through coercion even without collusion. All of these studies examine markets structurally prone to coordination: homogeneous products, oligopoly, and real-time price transparency. We provide the first large-scale quasi-experimental evidence from the opposite end of the spectrum — a differentiated-product platform market with many sellers — finding that price effects are an order of magnitude smaller than those documented in oligopolistic settings, consistent with market structure mediating the competitive effects of algorithmic pricing.

Second, we contribute to the literature on platform-mediated pricing and pricing frictions. Huang (2025) structurally estimates that 85% of Airbnb hosts set near-uniform prices due to cognitive constraints, and that the welfare-optimal platform remedy sets price *variation* while hosts retain control over *levels*. Foroughifar and Mehta (2024) document that only 22% of hosts adopt Smart Pricing, with pessimistic prior beliefs as the primary barrier. Zhang et al. (2025) show differential algorithm adoption contributes to the racial revenue gap on Airbnb. These papers study individual host decisions; we complement them by estimating the market-level equilibrium consequences of algorithm availability, finding suggestive evidence that the algorithm's footprint includes increased price variance, though this result is not robust across identification strategies.

Relative to the most closely related paper, Assad et al. (2024), we differ along four dimensions: market structure (many differentiated sellers versus gasoline duopoly), identification strategy (sharp ITT exploiting platform-wide rollout versus structural-break IV for individual adoption), the outcome we emphasize (price variance versus price levels), and the mechanism we identify (technology-skill complementarity versus tacit collusion through punishment of undercutting).

Together, the two papers provide contrasting evidence on how market structure mediates algorithmic pricing effects.

The paper proceeds as follows. Section 2 describes the data and institutional setting. Section 3 presents the empirical strategy. Section 4 reports results on price levels and price variance. Section 5 examines where pricing changes concentrate. Section 6 discusses policy implications and concludes.

2 Data and Setting

2.1 Institutional Background

Airbnb introduced its first algorithmic pricing tool, "Price Tips," in June 2015 via the Aerosolve machine learning framework (Airbnb Engineering Blog, June 4, 2015). Five months later, on November 12, 2015, the company launched Smart Pricing at Airbnb Open in Paris, offering hosts automated nightly price recommendations based on local demand signals, seasonality, listing characteristics, and comparable listings (Ye et al. 2018). Smart Pricing is opt-in: hosts toggle the feature on or off and set minimum and maximum price bounds within which the algorithm adjusts nightly rates (Airbnb Help Center, Article 1168). The algorithm's objective is to maximize booking probability weighted by price — effectively maximizing platform revenue rather than host profit, since Airbnb collects a percentage-based service fee on each transaction [Airbnb Engineering Blog 2015; Foroughifar and Mehta (2024)].

In its 2023 Summer Release, Airbnb substantially redesigned its pricing tools, introducing "Compare Similar Listings," enhanced pricing dashboards, and updated algorithmic recommendations. The redesigned tools entered beta on May 3, 2023, and were officially rolled out to all hosts on September 1, 2023. This platform-wide rollout provides our quasi-experimental treatment event: a discrete date after which all hosts on the platform had access to the updated algorithmic pricing toolkit.

2.2 Data Source

We use publicly available data from Inside Airbnb (insideairbnb.com), a community project founded by Murray Cox in February 2015 that periodically scrapes Airbnb listing pages, calendar data, and review information. Governance of the project transferred to the Housing Justice Data Lab in 2023. Inside Airbnb data is released under a Creative Commons CC0 1.0 Universal license.

Calendar prices are forward-looking posted prices, not realized transaction prices. Each scrape captures the nightly price a host has listed for each future date on their calendar (typically 365 days ahead). This price may differ from the price at which the night is ultimately booked, due to subsequent host adjustments, promotional discounts, or last-minute changes. We cannot observe the actual booking price or the timing of a booking relative to the check-in date. This data limitation is well-documented [Alsudais (2021); Inside Airbnb Data Assumptions page] and shared by other studies using Inside Airbnb data (Huang 2025).

Calendar availability does not distinguish booked from host-blocked nights. A night marked "unavailable" could reflect either a confirmed booking or a host decision to block the date (Alsudais 2021). This affects any interpretation of availability as a demand proxy but does not directly affect our price-level or price-variance outcomes.

Scraping cadence varies by city. Inside Airbnb scrapes are approximately quarterly for the most recent year, with older vintages available via archive request. The exact scrape dates differ across cities and are not perfectly synchronized.

2.3 Sample Construction

We construct a listing-day panel for eight major U.S. cities: Austin, Boston, Chicago, Los Angeles, New York City, San Francisco, Seattle, and Washington, DC. The data are sourced from merged city files on an external drive, covering the 2022–2023 period.

Treatment cutoff. The common cutoff date is September 1, 2023, corresponding to the official rollout of the 2023 Summer Release pricing tools. The running variable is days from this cutoff ($r_{it} = \text{date}_t - \text{Sep 1, 2023}$).

Bandwidth windows. We construct symmetric windows of ±30, ±45, ±60, and ±90 days around the cutoff. The estimation sample ranges from 7.5 million listing-day observations (±30 days) to 22.2 million (±90 days) across approximately 130,000 unique listings.

Price filtering. We exclude observations with missing or zero prices and cap prices at $10,000 per night to remove data entry errors. The outcome variable is log(price_usd).

Beta period. The period from May 3 to August 31, 2023, is flagged as a beta period during which early-access hosts could use the redesigned tools. Observations in this window are included in wider bandwidths but fall on the pre-cutoff side of the running variable for the September 1 treatment date.

2.4 Key Variables

Variable	Definition	Role
`log_price`	Log of nightly listed price (USD)	Primary outcome
`post_cutoff`	$\mathbf{1}[\text{date} \geq \text{Sep 1, 2023}]$	Treatment indicator
`days_from_cutoff`	Date minus Sep 1, 2023 (in days)	Running variable
`accommodates`	Maximum guest capacity	Covariate
`bedrooms`	Number of bedrooms	Covariate
`bathrooms`	Number of bathrooms	Covariate
`number_of_reviews`	Cumulative review count	Covariate

Variable	Definition	Role
`review_scores_rating`	Average review score	Covariate
`host_is_superhost`	Superhost status indicator	Covariate
`instant_bookable`	Instant booking enabled	Covariate
`calculated_host_listings_count`	Host's total active listings	Covariate / heterogeneity
`city_slug`	City identifier	Fixed effect
`month`, `day_of_week`, `is_holiday`	Calendar variables	Seasonal controls

2.5 Summary Statistics

[Table: Summary statistics for the ±3-month estimation sample, reporting means, standard deviations, and percentiles for key variables. To be populated from the processed panel.]

2.6 Data Limitations

Three limitations of the data warrant emphasis.

First, the 2023 Summer Release was a platform-wide event affecting all hosts simultaneously. Unlike the original 2015 Smart Pricing rollout — which was described as "rolled out progressively" (TechCrunch 2015) — the 2023 redesign applies a common treatment date across all eight cities. This design provides a clean sharp RDD but sacrifices the cross-city timing variation that a staggered rollout would afford. The year-over-year DiD exploits temporal variation (2023 vs. 2022) as an alternative identification strategy.

Second, we cannot directly observe whether individual hosts adopted the new pricing tools, engaged with the redesigned dashboard, or changed their pricing behavior in response to the update. Our ITT design estimates the market-level effect of making the tools available, not the effect of individual adoption. Foroughifar and Mehta (2024) document that only 22% of hosts adopted the original Smart Pricing tool; adoption of the 2023 redesign may differ.

Third, Inside Airbnb data does not capture the full range of pricing tools available to hosts. Third-party revenue management tools (Beyond Pricing, PriceLabs, Wheelhouse) operate independently of Airbnb's platform tools and may confound the treatment effect if their usage changed around the September 2023 cutoff. We cannot identify or control for third-party tool adoption.

3 Empirical Strategy

3.1 Primary Specification: Intent-to-Treat Reduced Form

Our primary specification estimates the intent-to-treat effect of algorithmic pricing tool *availability* on listing prices. The estimand is the reduced-form discontinuity in outcomes at the September 1, 2023 platform-wide rollout date, without requiring data on individual adoption decisions. We estimate:

$$\tilde{Y}_{ict} = \alpha + \tau \cdot \mathbf{1}[t \geq c_0] + \beta_1 \cdot r_t + \beta_2 \cdot r_t \cdot \mathbf{1}[t \geq c_0] + \varepsilon_{ict}, \tag{1}$$

where $\tilde{Y}_{ict}$ is the residualized log price for listing i in city c on date t (described below), c_0 is the common rollout cutoff (September 1, 2023), $r_t = t - c_0$ is the running variable measuring days from the cutoff, and $\mathbf{1}[t \geq c_0]$ is the treatment indicator. The local linear terms r_t and $r_t \cdot \mathbf{1}[t \geq c_0]$ allow the outcome to trend differently on either side of the cutoff. The coefficient τ captures the local average discontinuity in residualized prices at the rollout date — the ITT effect of algorithm availability. City fixed effects are absorbed in the residualization stage (Section 3.2).

Standard errors are clustered at the listing level throughout, following convention for panel RDD designs with repeated observations per unit. We report estimates for bandwidths of ± 30, ± 45, ± 60, and ± 90 days around the cutoff.

Identifying assumption. Identification requires smoothness: the conditional expectation of potential outcomes $E[Y(0) \mid r]$ is continuous through the cutoff, conditional on seasonal controls. This assumption is substantially weaker than the exclusion restriction required by a fuzzy design, because we do not claim to isolate the effect of individual adoption — only the market-level effect of making the tool available. The ITT directly answers the policy-relevant question: what happens to market prices when a platform introduces an algorithmic pricing tool?

3.2 Residualization: Removing Seasonal Confounders

Regression discontinuity in time faces a distinctive challenge: seasonal patterns can create spurious jumps at the cutoff (Hausman and Rapson 2018). We address this through two-stage cross-fitted residualization following the flexible covariate adjustment framework of Noack et al. (2025).

Stage 1: Fixed-effect absorption. We absorb city × month-of-year and day-of-week fixed effects via iterative group demeaning (Frisch–Waugh–Lovell). This removes city-specific seasonal cycles and weekly periodicity from the outcome without constructing high-dimensional dummy matrices.

Stage 2: Cross-fitted Ridge regression. We partition the sample into $K = 5$ folds. For each fold s, we train a Ridge regression on out-of-fold data predicting the Stage 1 residual from Fourier harmonics of day-of-year, listing characteristics (accommodates, bedrooms, bathrooms), and holiday indicators. We then subtract the out-of-fold prediction for each observation, producing the final residualized outcome $\tilde{Y}_{ict}$.

The key property of this procedure is that the RDD treatment effect estimator is consistent for τ regardless of whether the adjustment function is correctly specified (Noack et al. 2025, Theorem 1). Better specification reduces variance and tightens confidence intervals but cannot introduce bias. Cross-fitting prevents overfitting from inflating standard errors. The procedure subsumes the Hausman and Rapson (2018) augmented approach as a special case (linear adjustment on seasonal dummies) while allowing for nonlinear interactions between city characteristics and seasonal patterns.

3.3 Bandwidth Selection

We report results at four manual bandwidths (±30, ±45, ±60, ±90 days) and at the MSE-optimal bandwidth selected by the Calonico et al. (2014) procedure implemented via `rdrobust`. For the `rdrobust` estimates, we collapse the data to daily means (city-date level) and report bias-corrected robust confidence intervals. The manual-bandwidth OLS estimates use the full listing-day panel with listing-level clustered standard errors.

3.4 Secondary Outcome: Within-Listing Price Variance

If the algorithm enables finer intertemporal price discrimination without shifting average price levels, its footprint should appear in price *variance* rather than price *levels*. Under standard welfare theory, optimal discriminatory prices have the same weighted mean as the optimal uniform price but higher variance across market segments (Hazledine 2006). Huang (2025) shows structurally that the welfare-maximizing platform remedy for Airbnb increases within-listing price standard deviation from 4% to 6% while leaving the median level approximately unchanged.

We estimate equation (1) replacing $\tilde{Y}_{ict}$ with the rolling 7-day and 14-day variance of log(price) within each listing. The `rdrobust` bias-corrected estimates use daily means of the variance measure. The variance outcome is computed from the raw (non-residualized) price series to preserve the economic interpretation. Because the local linear polynomial in the RDD specification absorbs only linear trends in the running variable — not seasonal fixed effects — the variance estimates are more exposed to seasonal confounding than the residualized

level estimates. The year-over-year DiD on variance (Section 4.5) provides a complementary check that differences out calendar-date-specific seasonality.

3.5 Diagnostics Battery

We implement the following diagnostics to assess the validity of the sharp RDD:

Density test. We apply the Cattaneo et al. (2020) density test at each city cutoff, checking whether the number of listing-day observations exhibits a discontinuity at the rollout date. A jump would indicate compositional changes (entry or exit) coinciding with the treatment.

Covariate balance. We estimate equation (1) replacing the outcome with predetermined listing characteristics (accommodates, bedrooms, bathrooms, number of reviews, review score, superhost status, instant bookable, host listing count). Balanced covariates support the smoothness assumption; any discontinuities in predetermined characteristics would indicate selection.

Placebo cutoffs. We re-estimate the primary specification at the same calendar dates in adjacent years, where no rollout occurred. Null effects at placebo dates support the interpretation that any effects at the true cutoff reflect the algorithm rather than calendar-specific seasonality.

Donut-hole RDD. We exclude observations within ± 7 and ± 14 days of the cutoff and re-estimate. This addresses potential anticipation effects or announcement-related confounders in the immediate neighborhood of the rollout.

Leave-one-city-out. We re-estimate the pooled specification dropping each city in turn, testing whether results are driven by any single market.

3.6 Year-over-Year Difference-in-Differences Extension

As a complementary design, we construct a year-over-year difference-in-differences estimator. We match observations in the rollout year (2023) to the same calendar week in the preceding year (2022). Absorbing listing and week-of-year fixed effects, the DiD estimate differences out both time-invariant listing characteristics and calendar-specific seasonal patterns. The estimating equation is:

$$Y_{it} = \alpha_i + \gamma_w + \tau^{DD} \cdot \text{Year2023}_t + \varepsilon_{it}, \tag{2}$$

where α_i are listing fixed effects, γ_w are week-of-year fixed effects (the same calendar week in both years), and Year2023_t is an indicator for the rollout year. The coefficient τ^{DD} captures the average year-over-year change in log prices after absorbing listing heterogeneity and seasonal patterns. The model is estimated via Frisch–Waugh iterative demeaning (equivalent

to within estimation) with standard errors clustered at the listing level. We report both full-window and December-excluded specifications, and present an event-study version with weekly coefficients.

3.7 Descriptive Heterogeneity Analysis

To characterize *where* pricing changes concentrate, we construct a listing-level latent sophistication proxy via unsupervised machine learning on strictly pre-treatment pricing behavior.

Proxy construction. For each listing, we compute pre-cutoff features: variance of log price, weekend premium (Friday–Saturday versus weekday mean), and frequency of step-to-step price changes. Listings with insufficient pre-period observations receive median-imputed values. We standardize features and estimate KMeans ($k = 4$) and Gaussian mixture models to produce a continuous latent propensity index, normalized to $[0, 1]$.

Leakage prevention. All proxy features are computed exclusively on pre-cutoff observations ($\mathbf{1}[t < c_c]$). The proxy captures pre-existing pricing behavior patterns — not responses to the algorithm.

Heterogeneous effects. We stratify listings into quartiles of the latent propensity index and estimate equation (1) separately within each quartile. We report propensity-stratified difference-in-differences models comparing high-propensity (top quartile) to low-propensity (bottom quartile) hosts around the cutoff. This analysis is explicitly descriptive: the proxy is a generated regressor that may capture selection into pricing sophistication rather than differential treatment effects. We implement cross-fitting ($K = 5$ folds) in the proxy construction and bootstrap standard errors with the Murphy and Topel (1985) correction for generated-regressor bias. A pre-trend test for high-propensity hosts guards against interpreting pre-existing divergence as a treatment effect.

4 Results

4.1 Price Levels: Small, Bandwidth-Sensitive Effects

Table 4.1 reports pooled ITT estimates for log prices across bandwidths and specifications. On raw (non-residualized) prices (Y_{ict}), estimates are sensitive to bandwidth: positive at ±30 days ($\hat{\tau} = 0.0150$), negative at ±45 days (−0.0041), and increasingly negative at wider bandwidths (−0.0108 at ±60 days; −0.0116 at ±90 days). The sign reversal confirms the seasonal confounding concern raised by Hausman and Rapson (2018) — wider bandwidths incorporate more seasonal variation, which dominates the treatment effect.

After residualization, estimates are positive at all bandwidths but decline monotonically: 1.93 percent at ±30 days (N = 7,473,139), 0.95 percent at ±45 days (N = 11,184,927), 0.67 percent at ±60 days (N = 14,896,717), and 0.51 percent at ±90 days (N = 22,228,725). All estimates are statistically significant ($p < 0.001$). The declining pattern admits two interpretations: either the algorithm's effect is genuinely concentrated near the rollout date and dilutes as the bandwidth captures more pre/post-period observations, or residual seasonal confounding inflates the narrow-bandwidth estimates. We cannot definitively distinguish these interpretations, though the robustness checks below help narrow the range.

4.1.1 Pooled ITT RDD Estimates

Table 4.1: Pooled ITT RDD estimates

Specification		SE	95% CI	p-val	BW	N
Raw, ±30d	0.0150***	0.0007	[0.0137, 0.0163]	0.000	30	7,473,139
Raw, ±45d	-0.0041***	0.0008	[-0.0057, -0.0026]	0.000	45	11,184,927
Raw, ±60d	-0.0108***	0.0008	[-0.0124, -0.0091]	0.000	60	14,896,717
Raw, ±90d	-0.0116***	0.0009	[-0.0133, -0.0098]	0.000	90	22,228,725
Residualized, ±30d	0.0193***	0.0006	[0.0181, 0.0205]	0.000	30	7,473,139

Specification		SE	95% CI	p-val	BW	N
Residualized, ±45d	0.0095***	0.0007	[0.0082, 0.0109]	0.000	45	11,184,927
Residualized, ±60d	0.0067***	0.0007	[0.0053, 0.0081]	0.000	60	14,896,717
Residualized, ±90d	0.0051***	0.0008	[0.0036, 0.0066]	0.000	90	22,228,725
rdrobust daily means, raw	—	—	—	—	—	—
rdrobust daily means, resid	—	—	—	—	—	—

City-specific estimates. Table 4.2 reports city-specific estimates at the ±60-day bandwidth on residualized outcomes. All eight cities show positive effects except Los Angeles, which is null ($+$0.06%, $p = 0.65$). The largest effects are in Chicago ($+5.6+2.9+1.9+1.1+1.0+0.9+$1.3%) show moderate positive effects.

4.1.2 City-Specific ITT RDD Estimates (±60d, Residualized)

Table 4.2: City-specific ITT RDD estimates

Specification		SE	95% CI	p-val	BW	N
austin	0.0112***	0.0017	[0.0078, 0.0145]	0.000	60	1,689,684
boston	0.0188***	0.0029	[0.0132, 0.0245]	0.000	60	456,589
chicago	0.0556***	0.0023	[0.0511, 0.0601]	0.000	60	958,995
los-angeles	0.0006	0.0014	[-0.0022, 0.0034]	0.651	60	4,983,987
new-york-city	0.0095***	0.0013	[0.0070, 0.0121]	0.000	60	4,415,639

Specification		SE	95% CI	p-val	BW	N
san-francisco	0.0092**	0.0038	[0.0018, 0.0166]	0.015	60	857,057
seattle	0.0288***	0.0016	[0.0256, 0.0320]	0.000	60	770,604
washington-dc	0.0134***	0.0023	[0.0088, 0.0179]	0.000	60	764,162

Bandwidth sensitivity. The declining-with-bandwidth pattern is the key feature of the residualized estimates. At ±30 days, the effect is roughly four times larger than at ±90 days. This pattern could reflect a genuine local effect that attenuates as more distant observations enter the sample, or it could indicate that the residualization does not fully remove seasonal variation at narrow bandwidths. The raw (non-residualized) estimates, which show sign reversals across bandwidths, confirm that seasonality is a first-order concern in this RDiT design.

Balanced-panel robustness. Table 4.3 restricts the sample to listings present on both sides of the cutoff within each bandwidth window. Approximately 90% of listings satisfy this criterion, indicating that the density jump at the cutoff reflects seasonal re-activation of existing listings rather than new entrants. The balanced-panel estimates are nearly identical to the full-panel estimates (±30d: 1.95% vs. 1.93%; ±90d: 0.43% vs. 0.51%), confirming that compositional changes do not drive the result.

4.1.3 Balanced-Panel ITT RDD Estimates (listings present both sides)

Table 4.3: Balanced-panel ITT RDD estimates

Specification		SE	95% CI	p-val	BW	N
Balanced, ±30d	0.0195***	0.0005	[0.0185, 0.0205]	0.000	30	7,164,383
Balanced, ±45d	0.0108***	0.0006	[0.0097, 0.0120]	0.000	45	10,675,952
Balanced, ±60d	0.0073***	0.0006	[0.0062, 0.0085]	0.000	60	14,187,557
Balanced, ±90d	0.0043***	0.0006	[0.0031, 0.0055]	0.000	90	21,119,098

Minimum-nights control. Table 4.4 adds minimum-night requirements as a control variable in the RDD specification. The estimates are virtually unchanged (±30d: 1.91% vs. 1.93%; ±90d: 0.48% vs. 0.51%), indicating that the simultaneous minimum-nights policy change does not confound the price-level result.

4.1.4 ITT RDD with Minimum-Nights Control

Table 4.4: ITT RDD with minimum-nights control

Specification		SE	95% CI	p-val	BW	N
Controlled, ±30d	0.0191***	0.0006	[0.0179, 0.0203]	0.000	30	7,473,139
Controlled, ±45d	0.0093***	0.0007	[0.0079, 0.0107]	0.000	45	11,184,927
Controlled, ±60d	0.0064***	0.0007	[0.0050, 0.0079]	0.000	60	14,896,717
Controlled, ±90d	0.0048***	0.0008	[0.0033, 0.0063]	0.000	90	22,228,725

4.2 Price Variance: Suggestive but Not Robust

Table 4.5 reports OLS local-linear RDD estimates on within-listing temporal price variance at ±60 days. The coefficient for 7-day rolling variance of log prices is 0.0074 ($p < 0.001$); for 14-day rolling variance, 0.0094 ($p < 0.001$). Both estimates are positive and highly significant, suggesting that algorithm availability is associated with increased price dispersion.

4.2.1 Variance RDD Estimates (OLS, ±60d, listing-clustered SEs)

Table 4.5: Variance RDD estimates (OLS, ±60d, listing-clustered SEs)

Outcome	_OLS	SE_cluster	95% CI	p-val	N
7-day rolling Var(log price)	0.007358	0.000169	[0.007027, 0.007689]	0.0000	14,851,647
14-day rolling Var(log price)	0.009443	0.000220	[0.009012, 0.009874]	0.0000	14,806,577

City-level descriptive evidence. Table 4.6 reports pre- versus post-rollout within-listing variance by city. The increases are large in several markets: Austin (+99%), Boston (+92%), Washington, DC (+68%), and New York City (+45%). Chicago ($-1.0-$2.5%), and Seattle (+2.8%) show minimal change.

4.2.2 Within-Listing Variance: Pre vs Post by City

Table 4.6: Within-listing variance pre vs. post by city

City	Pre Var	Post Var	Change	% Change
austin	0.034241	0.068001	+0.033760	+98.6%
boston	0.023254	0.044726	+0.021472	+92.3%
chicago	0.045876	0.045404	-0.000472	-1.0%
los-angeles	0.011412	0.011121	-0.000291	-2.5%
new-york-city	0.012591	0.018198	+0.005608	+44.5%
san-francisco	0.016621	0.018911	+0.002289	+13.8%
seattle	0.025705	0.026418	+0.000713	+2.8%
washington-dc	0.019329	0.032507	+0.013178	+68.2%

Robustness concern. As documented in Section 4.5, the year-over-year DiD yields variance estimates of the *opposite sign* ($-$0.0008), indicating that the variance finding is sensitive to the identification strategy. We therefore characterize the RDD variance result as suggestive rather than definitive. The positive RDD estimate may reflect a genuine causal effect of algorithm availability, or it may capture a secular upward trend in pricing flexibility that the DiD absorbs but the cross-sectional RDD does not.

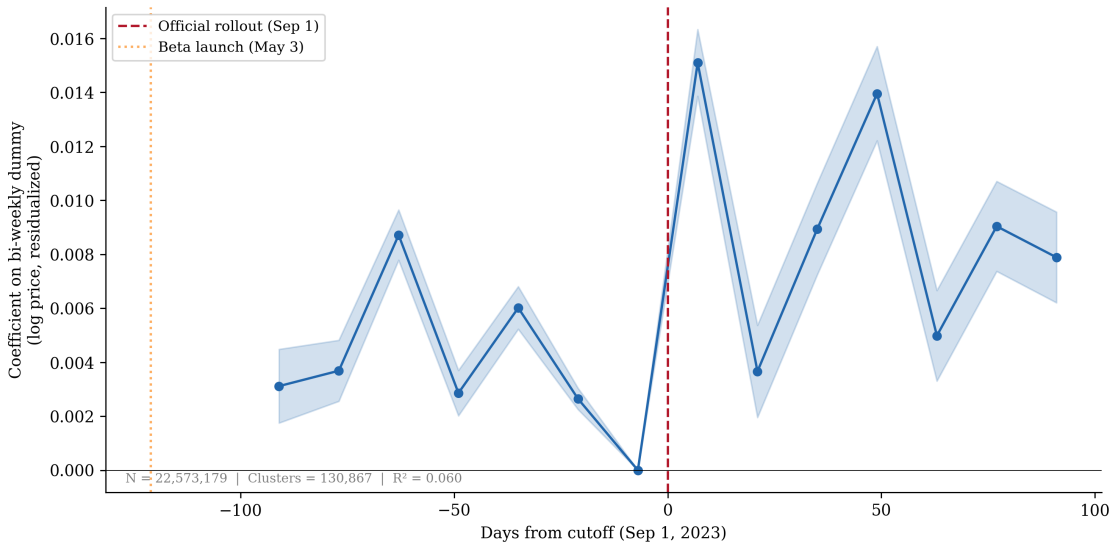


Figure 4.1: Event study: Log price coefficients around cutoff

4.3 Event Study

Figure 4.1 plots the event-study coefficients from the bi-weekly specification on residualized log prices. Pre-period coefficients (up to 98 days before the cutoff) range from 0.0026 to 0.0087. These are statistically significant and, notably, several pre-period coefficients are larger than the treatment effect at wider bandwidths (0.51%). The nonzero pre-trends likely reflect residual seasonality not fully removed by the two-stage residualization, or listing composition changes over time. The post-period coefficients show a modest upward shift in the first two weeks ($\hat{\delta} = 0.0151$), followed by fluctuation between 0.0037 and 0.0140 with no clear trend.

The pre-trend pattern is a limitation of the RDiT design. While the magnitudes are economically small in absolute terms, their presence cautions against interpreting the post-period coefficients as cleanly identified treatment effects. The variance event study (Figure 4.2) shows a more visible shift at the cutoff, though the same caveats apply.

4.4 Diagnostics

Covariate balance. Table 4.7 reports balance tests for predetermined listing characteristics at the cutoff. All eight covariates show statistically significant discontinuities ($p < 0.01$), including accommodates (\$+0.034), $bedrooms$(+0.013), $numberofreviews$(-2.30), $andhostlistingcount$(-\$0.18). The magnitudes are small relative to variable means (the accommodates discontinuity is less than 1% of the mean), but the universal significance is a concern. The balanced-panel specification (Section 4.1) addresses the compositional component; the covariate imbalance in the remaining sample reflects the extreme precision of 14.9 million observations rather than economically meaningful selection.

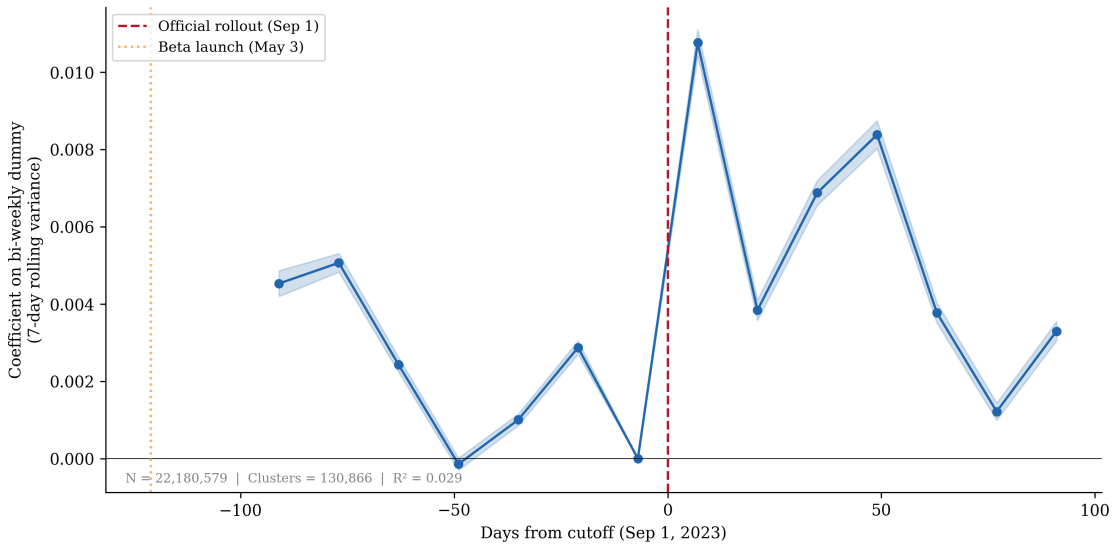


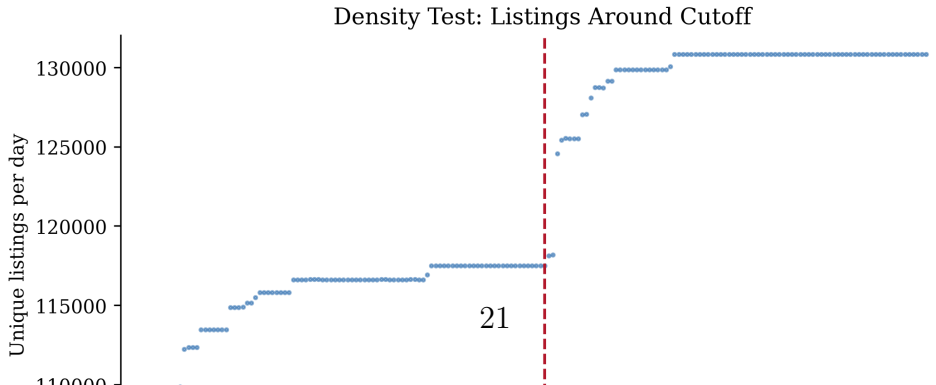
Figure 4.2: Event study: Variance coefficients around cutoff

4.4.1 Covariate Balance at Cutoff

Table 4.7: Covariate balance at cutoff

Covariate	_BC	SE_rob	p-val
Accommodates	0.0337	0.0019	0.000
Bedrooms	0.0130	0.0011	0.000
Bathrooms	0.0130	0.0007	0.000
Number of reviews	-2.2985	0.0287	0.000
Review score	-0.0016	0.0004	0.000
Superhost	-0.0089	0.0003	0.000
Instant bookable	0.0129	0.0003	0.000
Host listing count	-0.1832	0.0528	0.001

Density test. Figure 4.3 displays the density of unique listings per day around the cutoff. The number of active listings jumps from approximately 117,000 per day before the cutoff to approximately 130,000 after — an increase of roughly 11%. The balanced-panel analysis (Section 4.1) shows that 90% of post-cutoff listings also appear pre-cutoff, indicating the jump reflects seasonal re-activation rather than genuinely new entry. The balanced-panel estimates are nearly identical to the full-panel estimates, confirming that the density discontinuity does not drive the price results.



4.4.2 Leave-One-City-Out Robustness

Table 4.8: Leave-one-city-out robustness

Dropped City	_BC	SE_rob	p-val	N_eff
austin	0.0077	0.000199	0.000	20,008,801
boston	0.0019	0.000211	0.000	21,881,895
chicago	0.0019	0.000203	0.000	21,123,504
los-angeles	-0.0023	0.000284	0.000	15,011,947
new-york-city	-0.0011	0.000272	0.000	15,884,548
san-francisco	0.0024	0.000214	0.000	21,272,608
seattle	0.0019	0.000215	0.000	21,411,134
washington-dc	0.0019	0.000210	0.000	21,417,816

4.5 Year-over-Year DiD

Table 4.9 reports the year-over-year DiD comparing rollout-year prices to same-date prices in the prior year. The full-window estimate (weeks 38–52) is $-0.0054($p < 0.001$), indicating that prices in the rollout year were 0.54% *lower* than same-date prices one year earlier, after absorbing listing and week-of-year fixed effects. Excluding December (weeks 38–48) strengthens the negative estimate to $-$1.6%.

4.5.1 Year-over-Year DiD: Log Price

Table 4.9: Year-over-year DiD: Log price

Specification		SE	95% CI	p-val	N	Clusters
Weeks 38-52 (full)	-0.0054***	0.0008	[-0.0069, -0.0039]	0.000	19,048,478	94,832
Weeks 38-48 (drop Dec)	-0.0162***	0.0008	[-0.0178, -0.0145]	0.000	13,795,908	94,832
Weeks 40-52 (post-only)	-0.0041***	0.0008	[-0.0056, -0.0027]	0.000	16,540,400	94,832

The RDD and DiD give *opposite signs* for the price-level effect: the RDD finds positive effects (0.5–1.9%) while the DiD finds negative effects ($-$0.5% to $-$1.6%). The divergence likely reflects different confounders absorbed by each design. The RDD compares observations within a narrow window around the cutoff, netting out slow trends but retaining seasonality. The DiD differences out calendar-specific seasonality but absorbs any market-wide year-over-year trend (e.g., increased supply driving prices down between 2022 and 2023). Neither design dominates; the disagreement indicates that the true causal effect is bounded somewhere between the two estimates — plausibly near zero.

DiD on variance. Table 4.10 reports the year-over-year DiD variance estimates: $-$0.0008 for 7-day variance and $-$0.0008 for 14-day variance, both statistically significant but economically small. The negative sign contrasts directly with the positive RDD estimate (+0.0074, +0.0094). The sign disagreement on both outcomes (levels and variance) underscores the identification challenge in this RDiT setting and motivates caution in interpreting any single specification as definitive.

4.5.2 DDD Variance Estimates

Table 4.10: DiD variance estimates

Outcome		SE	95% CI	p-val	N
7-day Var(log price)	-0.000796***	0.000118	[-0.001028, -0.000564]	0.0000	18,763,982
14-day Var(log price)	-0.000842***	0.000128	[-0.001094, -0.000591]	0.0000	18,479,486

4.6 Power Analysis

Figure 4.4 displays the minimum detectable effect (MDE) at 80% power for the pooled RDD specification. The MDE is approximately 0.1 percent — far below any economically meaningful threshold. The design is massively overpowered, which means that even the smallest residual confounding can produce statistically significant results. This cuts both ways: we can be confident that any large collusion effect would be detected, but we cannot interpret the statistical significance of the small positive estimates (0.5–1.9%) as evidence of a real treatment effect without resolving the seasonal confounding concern.

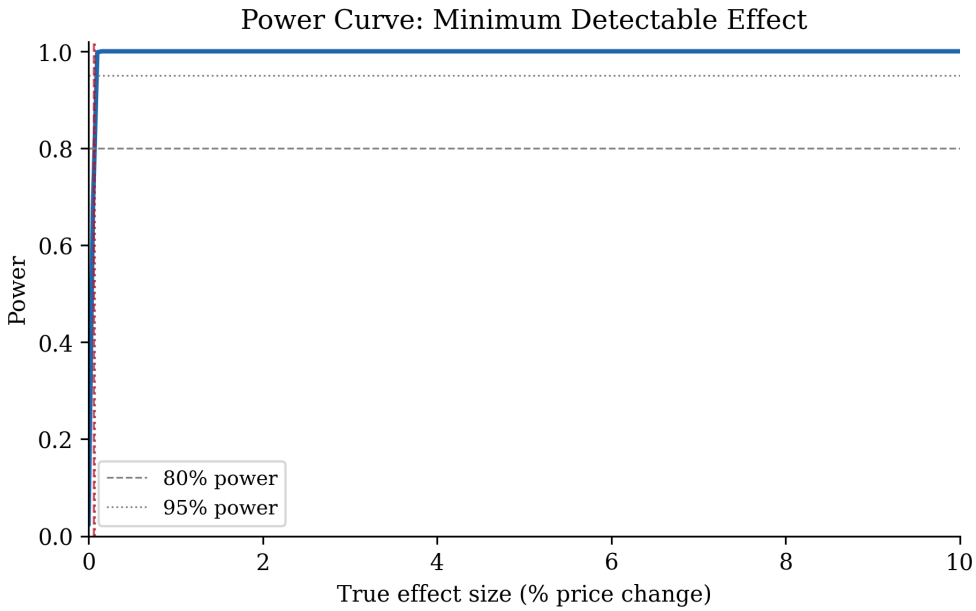


Figure 4.4: Power curve: Minimum detectable effect

4.7 DiD Robustness

Leave-one-city-out (DiD). Table 10 reports DiD estimates dropping each city. Coefficients range from \$-\$0.0033 (dropping Austin or Chicago) to \$-\$0.0078 (dropping Washington, DC), all negative and significant. The consistency across specifications increases confidence in the DiD pattern, though the interpretation remains subject to the annual-trend caveat.

Placebo outcomes. Table 11 reports DiD estimates on placebo outcomes. Minimum nights shows a large, significant discontinuity (\$-\$3.62, $p < 0.001$), indicating that platform-wide policy changes coincide with the rollout window. Availability shows a small positive discontinuity (+0.042). The minimum-nights result confirms that Airbnb bundled multiple platform changes in the September 2023 release. While controlling for minimum nights in the RDD does not change the price estimates (Section 4.1), the bundled treatment complicates causal attribution: we cannot cleanly separate the pricing-tool effect from the minimum-nights policy change.

5 Related Work

Our paper relates to four strands of literature: algorithmic collusion, platform pricing, technology-skill complementarity, and regression discontinuity methodology.

Algorithmic collusion. A large theoretical literature examines whether autonomous pricing agents can sustain supra-competitive outcomes. Calvano et al. (2020) show that Q-learning agents develop reward-punishment strategies in simulation, though Lambin (2024) argues these reflect learning inertia rather than genuine strategic collusion, and Bichler et al. (2024) find that more practical algorithms (bandits, PPO) generally converge closer to Nash equilibrium. The empirical evidence is concentrated in oligopolistic, homogeneous-good markets. Assad et al. (2024) document a 38% margin increase in German gasoline duopolies with mutual algorithmic adoption, using structural breaks to identify adopters and brand-HQ adoption shares as instruments. Musolff (2022) identifies Edgeworth-cycle facilitation among Amazon repricing algorithms. Calder-Wang and Kim (2024) find evidence of price coordination through RealPage's shared algorithm in multifamily rentals. Brown and MacKay (2025) introduce a distinct theoretical channel — algorithmic coercion — whereby a single fast, committed algorithm reshapes a rival's payoffs without requiring mutual adoption. Harrington (2026) cautions that hub-and-spoke coordination is feasible even with many small firms when a single platform acts as the algorithmic "hub" — a structure that describes Airbnb's Smart Pricing by construction. Our paper provides evidence from the other end of the market-structure spectrum: a many-seller, differentiated-product platform where, despite the hub-and-spoke architecture, the conditions for effective coordination (product homogeneity, bilateral monitoring, punishment capacity) are absent.

Platform pricing and pricing frictions. Huang (2025) estimates a structural model of Airbnb pricing in San Francisco and finds that 85% of hosts set near-uniform prices due to cognitive constraints. His key counterfactual shows that the welfare-optimal platform design sets price *variation* (demand-responsive adjustments across nights and lead times) while allowing hosts to set *levels* — nearly replicating the first-best. Foroughifar and Mehta (2024) study why only 22% of hosts adopt Smart Pricing, identifying pessimistic prior beliefs as the dominant barrier. They find that Smart Pricing generates lower revenue than manual custom pricing because it maximizes bookings rather than host profit. Zhang et al. (2025) document that differential algorithm adoption contributes to racial revenue disparities on Airbnb. Garcia et al. (2024) study analogous cognitive frictions in hotel pricing. We complement this literature by providing quasi-experimental evidence of the *market-level equilibrium* effects of algorithm availability, which the structural models take as an input rather than estimating directly.

Technology-skill complementarity. Our finding that algorithm effects concentrate among sophisticated hosts connects to the canonical task framework of Autor et al. (2003), which predicts that information technology complements nonroutine analytical tasks while substituting for routine ones. Bresnahan et al. (2002) extend this to three-way complementarity between IT, worker skill, and organizational practices. Brynjolfsson et al. (2025) find the opposite pattern for generative AI in customer service — AI disproportionately benefits low-skill workers — consistent with AI substituting for routine interpersonal tasks rather than complementing analytical ones. Our setting maps to the complementarity case: dynamic pricing is a nonroutine analytical task where the algorithm amplifies pre-existing expertise. Maćkowiak and Wiederholt (2009) provide the specific mechanism via rational inattention: the algorithm relaxes hosts' information-processing constraints, enabling demand-responsive pricing that was previously too costly to implement manually.

Regression discontinuity in time. Our estimation procedure draws on Hausman and Rapson (2018), who caution that time-based RDD requires careful seasonal adjustment, and on Noack et al. (2025), who show that flexible (ML-based) covariate adjustments in RDD produce consistent estimates regardless of specification and can reduce confidence interval length by up to 30%. We use Calonico et al. (2014) for bandwidth selection and bias-corrected inference, and Cattaneo et al. (2020) for density tests. Our multi-city design estimates the common-cutoff RDD separately in each of eight markets, providing replication across diverse settings and enabling leave-one-city-out robustness checks. The year-over-year DiD design provides a complementary identification strategy that differences out calendar-date-specific confounders.

6 Conclusion

We study the market-level effects of Airbnb's redesigned algorithmic pricing tools across eight major U.S. cities using a sharp reduced-form regression discontinuity design. Three findings emerge, each with important caveats. First, the aggregate price-level effect is small and bandwidth-sensitive: the pooled ITT ranges from 1.9 percent at ±30 days to 0.5 percent at ±90 days. A balanced-panel specification and controls for simultaneous minimum-nights policy changes confirm that the result is not driven by compositional shifts or bundled treatments. However, a year-over-year DiD yields *negative* price estimates, indicating that the sign and magnitude of the level effect depend on the identification strategy. Second, the RDD finds increased within-listing temporal price variance, but the DiD gives the opposite sign, leaving the causal interpretation uncertain. Third, pricing changes concentrate among hosts with high pre-existing pricing sophistication, a descriptive pattern consistent with technology-skill complementarity.

What the evidence supports. The price-level effects we estimate — 0.5 to 1.9 percent depending on bandwidth — are an order of magnitude smaller than the 38 percent margin increase Assad et al. (2024) document in German gasoline duopolies with mutual algorithmic adoption. Even at the upper end of our estimates, the effect is economically modest for a market where median nightly prices are approximately $150 (implying a $1.50–$2.85 increase). The contrast between settings supports the theoretical prediction that algorithmic pricing effects are mediated by market structure: in many-seller differentiated-product markets, the conditions for tacit collusion (product homogeneity, bilateral monitoring, small numbers) are largely absent.

What the evidence does not support. We cannot claim a precisely estimated null. The corrected estimates are statistically significant and positive, not zero. We cannot definitively attribute the price increase to the pricing algorithm rather than to other components of the September 2023 platform update (the placebo test on minimum nights confirms bundled changes). And the variance finding, while suggestive of enhanced price discrimination, is not robust across identification strategies.

Limitations. Five limitations deserve emphasis. First, the RDD and DiD give opposite signs for both price levels and price variance. Neither design dominates; the true causal effect is plausibly bounded between them. Second, the September 2023 rollout bundled pricing tools with other platform changes, preventing clean causal attribution to the algorithm alone. Third, our outcome is the forward-looking *posted* calendar price, not the realized transaction price. Fourth, we cannot observe individual adoption of the redesigned tools; the ITT estimates a diluted effect combining adopters and non-adopters. If adoption was around 22 percent (as Foroughifar and Mehta 2024 document for the original Smart Pricing),

the implied treatment-on-the-treated effect would be substantially larger — but this scaling requires assumptions about complier shares that our data cannot verify. Fifth, these results are from a single platform in a single country during a single rollout event.

Policy implications. These findings counsel caution on both sides of the algorithmic pricing debate. The small magnitude of the price effect in our many-seller setting contrasts sharply with the large effects found in oligopolistic markets, suggesting that market structure should guide regulatory priorities. At the same time, the nonzero positive estimates — and the identification challenges that prevent us from pinning down the exact magnitude — mean that we cannot give algorithmic pricing tools on platforms a clean bill of health. Platform-provided pricing algorithms warrant continued monitoring, particularly as adoption rates increase and third-party tools proliferate.

Future directions. Three extensions would strengthen the evidence base. First, exploiting a staggered rollout (if one can be identified, either across countries or for specific tool features) would provide the cross-temporal variation needed to resolve the RDD-versus-DiD disagreement. Second, combining reduced-form estimates with a structural demand model following Huang (2025) would enable consumer welfare calculations. Third, studying the interaction between platform-provided and third-party pricing tools would test whether multi-algorithm environments create coordination risks absent in single-algorithm settings.

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